

KUWAIT UNIVERSITY
Department of Mathematics

MATH 211: CALCULUS III

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1. [5 pts each] *In each part, choose the correct answer.*

i. What is the minimum value of the function $f(x, y) = 2x + 4y$ on the circle $x^2 + y^2 = 5$?

- (a) $-4\sqrt{5}$ (b) $-2\sqrt{5}$ (c) -20 (d) -10 (e) -6

ii. What is the value of the integral $\int_0^{\pi/4} \int_0^{\cos x} \sin x \, dy \, dx$?

- (a) 1 (b) $\frac{3}{4}$ (c) $\frac{3}{8}$ (d) $\frac{1}{2}$ (e) $\frac{1}{4}$

iii. What is the rate of change of the function $f(x, y, z) = 3xy^2 - 2yz$ at the point $P(3, 1, 4)$ in the direction from P to $Q(2, 3, 2)$?

- (a) 3 (b) 7 (c) 9 (d) 15 (e) 21

iv. Let $z = x^3y$ where $x = t^2$ and $y = 1 - 2t$. What is the value of $\frac{dz}{dt}$ at the point $(1, 3)$?

- (a) -20 (b) 20 (c) -16 (d) 16 (e) Undefined.

v. If C is any smooth curve with initial point $(1, 1)$ and terminal point $(2, 3)$, what is the value of the line integral $\int_C dx + dy$?

- (a) 1 (b) 2 (c) 3 (d) 5 (e) 0

vi. Let S be the surface with vector representation $\mathbf{r}(u, v) = \langle u + v, u - v, u^2 \rangle$. Which of the following is an equation for the tangent plane to S at the point $\mathbf{r}(1, 1) = (2, 0, 1)$?

- (a) $x + y + z = 1$ (b) $x + y - z = 1$ (c) $x - y - z = 1$
(d) $x + y + z = -1$ (e) $x - y - z = -1$

2. [15 pts] Let E be the solid that lies above the xy -plane and between the spheres $x^2 + y^2 + z^2 = 1$ and $x^2 + y^2 + z^2 = 4$. Find the mass of E if the density function is $f(x, y, z) = \sqrt{x^2 + y^2}$.

3. [15 pts] Let R be the region in the first quadrant enclosed by the hyperbolas $xy = 1$ and $xy = 4$, and the lines $y = x$ and $y = 3x$. Evaluate the double integral

$$\iint_R xy^3 \, dA$$

using the change of variables $x = \frac{v}{u}$, $y = uv$ where $u, v > 0$.

4. [10 pts] Let S be the part of the paraboloid $z = x^2 + y^2$ that lies below the plane $z = 4$, with positive (upward) orientation. Compute the flux of the vector field $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ across S .

5. [10 pts] A wire takes the shape of the semicircle $x^2 + y^2 = 1$, $x \geq 0$. Find the center of mass of the wire if the linear density at any point is 1.

6. [10 pts] Let C be the trapezoid with vertices $(1, 0)$, $(3, 0)$, $(5, 2)$, and $(2, 2)$. Use Green's Theorem to show that

$$\oint_C (\sin x + e^{-y}) \, dx + (xe^y + y^3 + 2x) \, dy > 0.$$

7. [10 pts] Complete the following statement of the Fundamental Theorem for Line Integrals, and then prove this theorem:

Theorem. *Let C be a smooth curve given by the vector function $\mathbf{r}(t)$, $a \leq t \leq b$. Let f be a differentiable function of three variables whose gradient vector ∇f is continuous on C . Then ...*

1. i. The correct choice is (d) -10 .

Let $g(x, y) = x^2 + y^2$. By the method of Lagrange multipliers, the maximum occurs at a solution of the system

$$\begin{cases} \nabla f = \lambda \nabla g \\ g = 5 \end{cases} \iff \begin{cases} 2 = \lambda(2x) \\ 4 = \lambda(2y) \\ x^2 + y^2 = 5. \end{cases}$$

The first two equations imply $\lambda \neq 0$, $x = 1/\lambda$ and $y = 2/\lambda$. Substituting these in the last equation we obtain $\lambda = \pm 1$ which gives the two solutions $(1, 2)$ and $(-1, -2)$. The result follows by comparing the values $f(1, 2) = 10$ and $f(-1, -2) = -10$.

- ii. The correct choice is (e) $\frac{1}{4}$.

$$\int_0^{\pi/4} \int_0^{\cos x} \sin x \, dy \, dx = \int_0^{\pi/4} \sin x \cos x \, dx = \left[\frac{1}{2} \sin^2 x \right]_0^{\pi/4} = \frac{1}{4}.$$

- iii. The correct choice is (b) 7 .

We have

$$\nabla f = \langle 3y^2, 6xy - 2z, -2y \rangle \Rightarrow \nabla f(3, 1, 4) = \langle 3, 10, -2 \rangle$$

and

$$\overrightarrow{PQ} = \langle -1, 2, -2 \rangle \Rightarrow |\overrightarrow{PQ}| = 3.$$

Now the desired directional derivative equals

$$\nabla f(3, 1, 4) \cdot \frac{\overrightarrow{PQ}}{|\overrightarrow{PQ}|} = \frac{1}{3} \langle 3, 10, -2 \rangle \cdot \langle -1, 2, -2 \rangle = 7.$$

- iv. The correct choice is (a) -20 .

By the chain rule, $\frac{dz}{dt} = (3x^2y)(2t) + (x^3)(-2)$. On the other hand, at the point $(1, 3)$ we have $x = 1$ and $y = 3$, thus $t = -1$. Evaluating dz/dt at these values yields -20 .

- v. The correct choice is (c) 3 .

The vector field $\mathbf{F}(x, y) = \mathbf{i} + \mathbf{j}$ is conservative with potential function $f(x, y) = x + y$. Thus the given line integral is independent of path and evaluates to $f(2, 3) - f(1, 1) = 5 - 2 = 3$.

- vi. The correct choice is (b) $x + y - z = 1$.

We have $\mathbf{r}_u = \langle 1, 1, 2u \rangle$ and $\mathbf{r}_v = \langle 1, -1, 0 \rangle$. At $(u, v) = (1, 1)$ these yield

$$\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 2 \\ 1 & -1 & 0 \end{vmatrix} = \langle 2, 2, -2 \rangle.$$

Thus the desired tangent plane has the equation $2(x - 2) + 2(y - 0) - 2(z - 1) = 0$.

2. In spherical coordinates, E is described as $1 \leq \rho \leq 2$, $0 \leq \theta \leq 2\pi$, $0 \leq \phi \leq \pi/2$, while the density function is $\sqrt{x^2 + y^2} = \rho \sin \phi$. Using these we have

$$\begin{aligned} m &= \iiint_E f(x, y, z) \, dV = \int_0^{2\pi} \int_0^{\pi/2} \int_1^2 \rho \sin \phi \, \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = 2\pi \int_1^2 \rho^3 \, d\rho \int_0^{\pi/2} \sin^2 \phi \, d\phi \\ &= 2\pi \left[\frac{\rho^4}{4} \right]_1^2 \int_0^{\pi/2} \frac{1 - \cos 2\phi}{2} \, d\phi = \frac{15\pi}{4} \left[\phi - \frac{1}{2} \sin 2\phi \right]_0^{\pi/2} = \frac{15\pi^2}{8}. \end{aligned}$$

3. Let T denote the given transformation $x = \frac{v}{u}$, $y = uv$. Note that under T we have $xy = v^2$ and $y/x = u^2$. Thus the given curves in the xy -plane correspond to the curves $v = 1$, $v = 2$, $u = 1$ and $u = \sqrt{3}$ in the first quadrant of the uv -plane. Hence R is the image under T of the region

$$S = \{(u, v) \mid 1 \leq u \leq \sqrt{3}, 1 \leq v \leq 2\}.$$

On the other hand, the Jacobian of T is

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} -\frac{v}{u^2} & \frac{1}{u} \\ v & u \end{vmatrix} = -\frac{2v}{u} \Rightarrow |J| = \frac{2u}{v} \quad (\text{since } u, v > 0).$$

In turn we obtain

$$\iint_R xy^3 \, dA = \int_1^{\sqrt{3}} \int_1^2 \frac{v}{u} (uv)^3 \cdot \frac{2u}{v} \, dv \, du = 2 \int_1^{\sqrt{3}} u \, du \int_1^2 v^5 \, dv = 2 \left[\frac{u^2}{2} \right]_1^{\sqrt{3}} \left[\frac{v^6}{6} \right]_1^2 = 21.$$

4. The surface S may be parametrized by

$$\mathbf{r}(x, y) = \langle x, y, x^2 + y^2 \rangle, \quad \text{where } (x, y) \in D = \{(x, y) \mid x^2 + y^2 \leq 4\}.$$

Using this parametrization the desired flux is computed as

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} \, dS &= \iint_D \mathbf{F}(x, y, x^2 + y^2) \cdot (\mathbf{r}_x \times \mathbf{r}_y) \, dA \\ &= \iint_D \langle x, y, x^2 + y^2 \rangle \cdot \langle -2x, -2y, 1 \rangle \, dA \\ &= - \iint_D (x^2 + y^2) \, dA = - \int_0^{2\pi} \int_0^2 r^2 r \, dr \, d\theta = -2\pi \left[\frac{r^4}{4} \right]_0^2 = -8\pi. \end{aligned}$$

5. Let C denote the given semicircle, and $(\bar{x}, \bar{y})$ denote the center of mass of the wire. By symmetry of the wire and its density function about the x -axis, we have $\bar{y} = 0$.

On the other hand, C is represented by the vector function $\mathbf{r}(t) = \langle \cos t, \sin t \rangle$ where $t \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. Thus $|\mathbf{r}'(t)| = |(-\sin t, \cos t)| = 1$, and we have

$$m = \int_C 1 \, ds = \int_{-\pi/2}^{\pi/2} |\mathbf{r}'(t)| \, dt = \int_{-\pi/2}^{\pi/2} dt = \pi,$$

and

$$\bar{x} = \frac{1}{m} \int_C x \, ds = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \cos t |\mathbf{r}'(t)| \, dt = \frac{2}{\pi} \int_0^{\pi/2} \cos t \, dt = \frac{2}{\pi} \left[\sin t \right]_0^{\pi/2} = \frac{2}{\pi}.$$

6. The functions $P(x, y) = \sin x + e^{-y}$ and $Q(x, y) = xe^y + y^3 + 2x$ have continuous partial derivatives everywhere in $\mathbb{R}^2$ and C is a polygonal curve. Hence all conditions of Green's Theorem are satisfied and we have

$$\oint_C P(x, y) \, dx + Q(x, y) \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dA = \iint_D (e^y + 2 + e^{-y}) \, dA,$$

where D is the region in the xy -plane bounded by C . Now the result follows from the fact that $e^y + 2 + e^{-y} > 0$ for all $(x, y) \in D$, and that D has a nonzero area.

7. See pages 1087 and 1088 of the textbook.